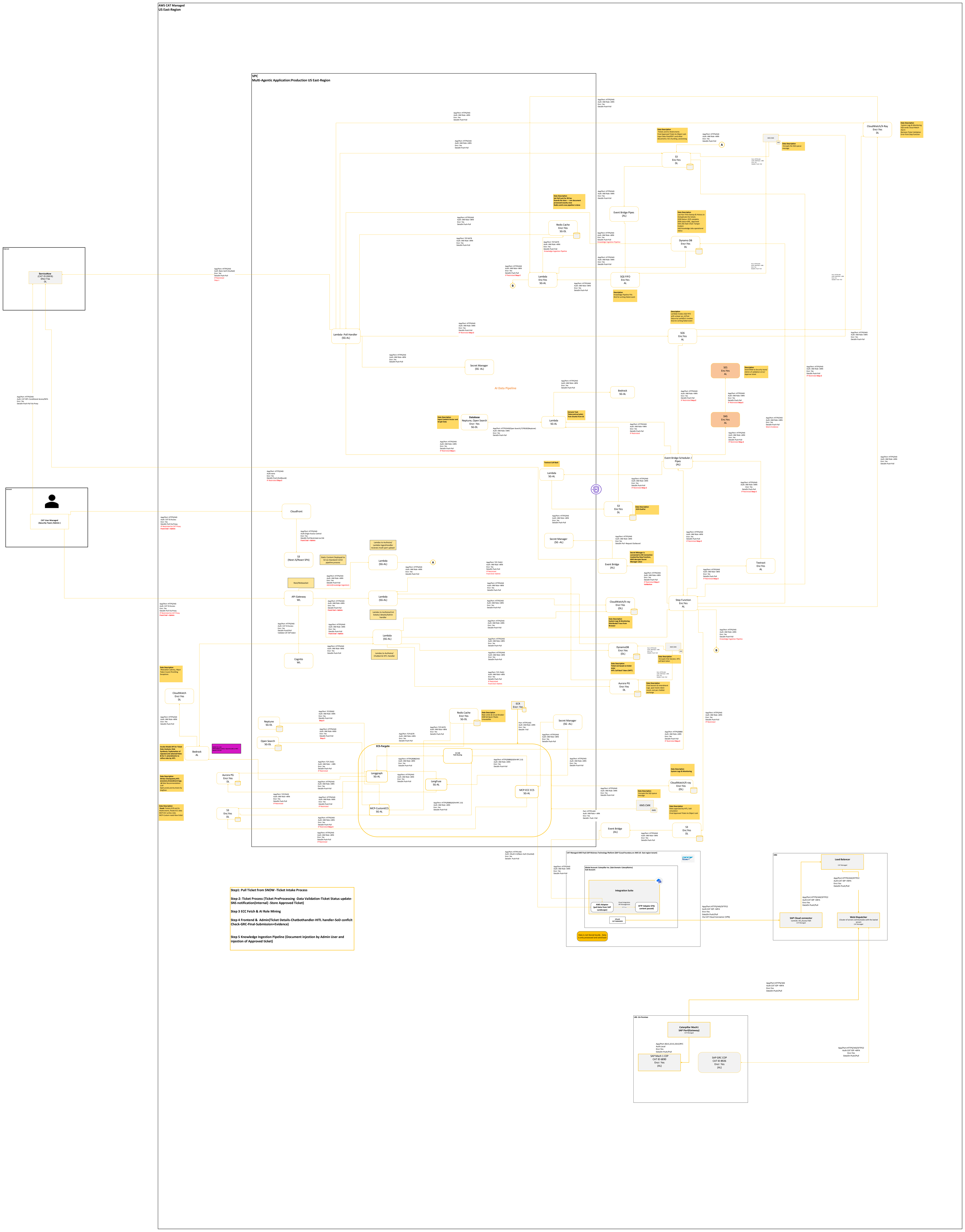
Agenctic AI for SAP Security
Caterpillar ID: Need To Create


AWS CAT Managed
US East-Region
VPC
Multi-Agentic Application:Production US East-Region
Internet
ServiceNow
(CAT ID:20816)
ENcr:Yes
DL
Intranet
CAT User Managed
(Security Team /Admin)
App/Port: HTTPS/443
Auth: Basic Auth (Vaulted)
Encr: Yes
DataDir:Push-Pull
IP Restricted
Step-1
App/Port: HTTPS/443
Auth: CAT IDP+ Conditional Access/MFA
Encr: Yes
DataDir:Push-Pull Via Proxy
Lambda- Poll Handler
(SG-AL)
App/Port: HTTPS/443
Auth: IAM Role +ARN
Encr: Yes
DataDir:Push-Pull
IP Restricted Step-2
Secret Manager
(SG -AL)
Redis Cache
Encr:Yes
SG-DL
App/Port: TCP:6379
Auth: IAM Role +ARN
Encr: Yes
DataDir:Push-Pull
Knowledge Ingestion Pipeline
Lambda
Enc:Yes
SG-AL
B
Event Bridge Pipes
(AL)
SQS FIFO
Enc:Yes
AL
S3
Enc:Yes
DL
A
KMS CMK
KMS
Port: HTTPS:443
Auth: IAM Role + ARN
Encr: Yes
DataDir: Push -Pull
Dynamo DB
Encr:Yes
DL
CloudWatch/X-Ray
Encr:Yes
DL
SQS
Enc:Yes
AL
SES
Enc:Yes
AL
SNS
Enc:Yes
AL
Worm Evidence
AI Data Pipeline
Bedrock
SG-AL
Database
Neptune, Open Search
Encr: Yes
SG-DL
App/Port: HTTPS/443(Open Search),TCP8182(Neptune)
Auth: IAM Role +ARN
Encr: Yes
DataDir:Push-Pull
IP Restricted Step-1
Lambda
SG-AL
Textract Call Back
App/Port: HTTPS/443
Auth:none
Encr: Yes
DataDir:Push (OutBound)
IP Restricted Step-2
Event Bridge Scheduler /
Pipes
(AL)
Secret Manager
(SG -AL)
App/Port: HTTPS/443
Auth: IAM Role +ARN
Encr Yes
DataDir:Pull -Request Outbound
Event Bridge
(AL)
IP Restricted Step-2
Validation
Textract
Enc:Yes
AL
Step Function
Enc:Yes
AL
App/Port: HTTPS/8060
Auth: IAM Role +ARN
Encr: Yes
DataDir:Push-Pull
IP Restricted Step-2
Cloudfront
App/Port: HTTPS/443
Auth: CAT ID Access
Encr: Yes
DataDir:Pull Via Proxy
IP Restricted to CAT Proxy
Front End + Admin
App/Port: HTTPS/443
Auth:Origin Access Control
Encr: Yes
DataDir:Pull Restricted via OAI
Front End + Admin
S3
(Next JS/React SPA)
Static Content Deployed to S3 via Standard CI/CD pipeline process
Rest/Websocket
API Gateway
WL
App/Port: HTTPS/443
Auth: CAT ID Access
Encr: Yes
DataDir:Push/Pull
Validate CAT IDP token
Cognito
WL
Lambda
(SG-AL)
Lamdas to Authoize/ Lambda IngestHandler recieves multi-part upload
Admin(Knowledge Ingestion)
Lamdas to Authoize/List tickets/ details/Admin handler
Lamdas to Authoize/ Chatbot & HITL Handler
App/Port: TCP /5432
Auth: IAM Role +ARN
Encr: Yes
DataDir:Push-Pull
IP Restricted
Front End +Admin
CloudWatch/X-ray
Encr:Yes
(DL)
DynamoDB
Encr:Yes
(DL)
Aurora PG
Encr:Yes
DL
CloudWatch
Encr:Yes
DL
Bedrock
AL
Neptune
SG-DL
Open Search
SG-DL
App/Port: TCP/8182
Auth: IAM Role +ARN
Encr: Yes
DataDir:Push-Pull
Step-3
Redis Cache
Encr:Yes
SG-DL
ECR
Encr: Yes
Port: HTTPS:443
Auth: IAM Role +ARN
Encr: Yes
DataDir: Pull
Secret Manager
(SG -AL)
ECS-Fargate
Int ALB
Path Routing
Langgraph
SG-AL
LangFuse
SG-AL
MCP-ECC ECS
SG-AL
MCP-CustomECS
SG-AL
App/Port: HTTPS/8080/443
App/Port: HTTPS/8080(JSON-RPC 2.0)
Aurora PG
Encr:Yes
DL
S3
Enc:Yes
DL
App/Port: TCP/5432
IP Restricted Step-3
CloudWatch/X-ray
Encr:Yes
(DL)
KMS CMK
KMS
Event Bridge
(AL)
S3
Enc:Yes
DL
App/Port: HTTPS:443
Auth: OAuth 2.0/Basic Auth (Vaulted)
Encr: Yes
DataDir: Push-Pull
CAT Managed AWS PaaS-SAP BUsiness Technology Platform (SAP CLoud Foundary on AWS US -East region tenant)
SAP
Global Account: Caterpillar Inc. (Sub-Domain: Caterpillarinc)
Sub Account:
Integration Suite
AWS Adaptor
(pull data from SAP Landscape)
Cloud Integration
API Management
SFTP Adaptor (File content passed)
(IPaaS)
CAT MANAGED
Data is not Stored locally . Data is only processed and send back
App/Port:HTTPS/443/SFTP22
Auth:CAT IDP +MFA
Encr:Yes
DataDir:Push/Pull
Via CAT Cloud Connector (VPN)
HRD
Load Balancer
CAT Managed
SAP Cloud connector
Controls: ACL,Access Path
CAT Managed
Web Dispatcher
(Cluster of servers communicates with the bacled server)
CAT Managed
App/Port:HTTPS/443
Auth:CAT IDP +MFA
Encr:Yes
DataDir:Push/Pull
LRD -On Premises
Caterpillar Mach1
SAP Fiori(Gateway)
CAT Managed
App/Port:3615,3215,3315/RFC
Auth:Local
Encr:Yes
DataDir:Push/Pull
SAP Mach 1 CEP
CAT ID 6890
Encr: Yes
(AL)
SAP GRC COP
CAT ID 8926
Encr: Yes
(AL)

Data Description
Tickets and its Attahcments
Final Approved Ticket As Object Lock
Input Data Files(PDF's and other documents) for chunking ,vectorizing

Data Description
Encrypts the SQS queue message

Data Description
System Logs & Monitoring
SQS sends Cloud Watch Alarm
Recieves Ticket Validation Error from Step function

Data Description
Set Poll Lock for 90 Sec
Guards the door — one document processed exactly once
Redis comit once pipeline is done

Data Description
Last Run Time Stamps & History to Deduplicate the tickets
DDB Status= OCR complete
DDB status=GRC_Approved
SHA-256 Hash Chain Tamper Evident
Add Knowledge Jobs-operational status

Description
Knowledge Pipeline Fifo
DLQ for writing failed event

Description
Lambda invokes SQS FIFO with unique sys_id.Post failure by Lambda it invokes DLQ for writing failed event

Description
Send EMail to Security team/ Admin of validation error/ Approve ticket

Data Description
Input Content Vector and Graph Data

Extracts Task Token,extract plain Text-chunks from S3

Data Description
OCR OutPut

Secret MAnager is connected to EB Connection invoked by Step Function, KMS decrypts Secret Manager value

Data Description
System Logs & Monitoring Distributed Trace from Browser

Data Description
Ticket List based on ticket state
HITL Call Back Token (WFT)

Data Description
Encrypts Chat Session, HITL call back token

Data Description
Chat Session & Amendment Logs ,span tracks token count, cost per chatbot exchange

Data Description
iNvocation Latency, INput Token Count,Throttling Exceptions

Invoke Model API for Ticket Data Analysis, Role Synthesis, Explaination of rejected and selected Roles & for 3 -amendments to refine roles by HITL

Data Description
Writes Checkpoints,HITL sessions.Amendment logs.
AI Role Recommendation LLM
Spans,evals,scores,traces by langfuse

Data Description
Reads Tickets,OCR and its Attahcments, Reads ECC data
MCP-ECC writes roles
MCP-Custom reads Raw ticket

Data Description
Rate Limits & Circuit Breaker DDB fall Back If Redis Unavailable

Data Description
System Logs & Monitoring

Data Description
Roles approved by HITL, SoD simulation
Final Approved Ticket As Object Lock

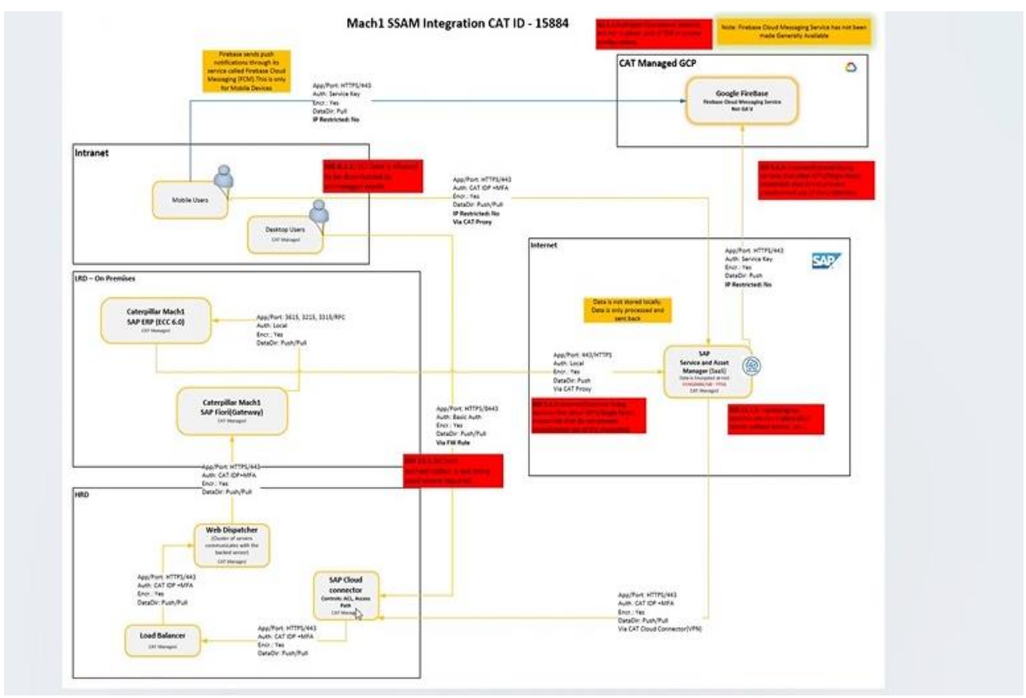
Step1: Pull Ticket from SNOW -Ticket Intake Process

Step-2: Ticket Process (Ticket PreProcessing -Data Validation-Ticket Status update-SNS notification(Internal) -Store Approved Ticket)

Step 3 ECC Fetch & AI Role Mining

Step 4 Frontend & Admin(Ticket Details-Chatbothandler-HITL handler-SoD conflcit Check-GRC-FInal-Submission+Evidence)

Step 5 Knowledge Ingestion Pipeline (Document injestion by Admin User and injestion of Approved ticket)


Mach1 SSAM Integration CAT ID - 15884